



LockBench

An automotive-radar chirp-PLL loop-filter designer

PVT-robustness verification · machine-learning inverse design

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Live: controlbench.onrender.com

THE PROBLEM

The frequency chirp is made by a PLL

Every FMCW automotive radar sweeps a frequency chirp and listens to the echo to measure range and speed.

That chirp is generated by a phase-locked loop (PLL).

The one part an engineer hand-tunes is the PLL loop filter — a resistor and two capacitors.

It sets chirp settling, phase noise and stability — and must hold from $-40\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$.

The chirp signal chain

Reference oscillator

Phase detector + charge pump

Loop filter (R, C_1, C_2) ← design

VCO $\div N$

$\times 8$ multiplier

77 GHz radar transmit

From requirements to a buildable, qualified design

1

Pick a real chip

LMX2594 · ADF5355 · ADF4351 —
datasheet parameters

2

Set radar targets

loop bandwidth, phase margin,
temperature grade

3

Design + verify

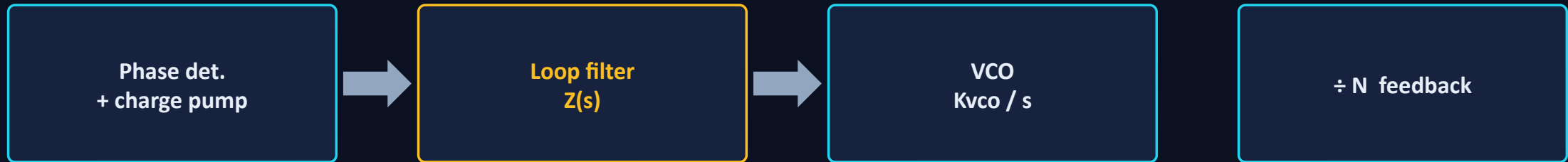
synthesise R/C, then check every
PVT corner

4

Get the result

buildable values + Qualified verdict
+ charts

A PLL is a feedback control loop



$$L(s) = [I_{cp} \cdot K_{vco} / (2\pi N)] \cdot Z(s) / s$$

The VCO's $1/s$ plus the filter's pole at the origin make it a Type-II loop, so the loop filter's zero is what recovers phase margin.

This is the linear charge-pump model of Gardner (1980) — the exact math LockBench implements.

How a design is produced

1

Build the loop model

from the real chip (I_{cp} , K_{vco} , N) + candidate filter

2

Synthesise the filter

solve R , C_1 , C_2 from bandwidth + phase-margin targets

3

Measure, don't assume

phase margin, chirp settling, jitter peaking from $L(s)/H(s)$

4

Sweep temperature corners

perturb K_{vco}/I_{cp} per AEC-Q100 grade; flag any violation

5

Refer to 77 GHz

$\times N$ multiplier: $+20 \cdot \log_{10} N$ phase noise, jitter invariant

6

Return the result

R/C values, verdict, per-corner metrics, plots

What makes it stand out



Real silicon

Anchored on datasheet parameters of shipping synthesizer ICs — not toy numbers.



Temperature-robust

Proves the design across AEC-Q100 corners, not just at 25 °C. The core contribution.



Honest ML

Inverse design that accelerates the physics model and is always re-verified by it.

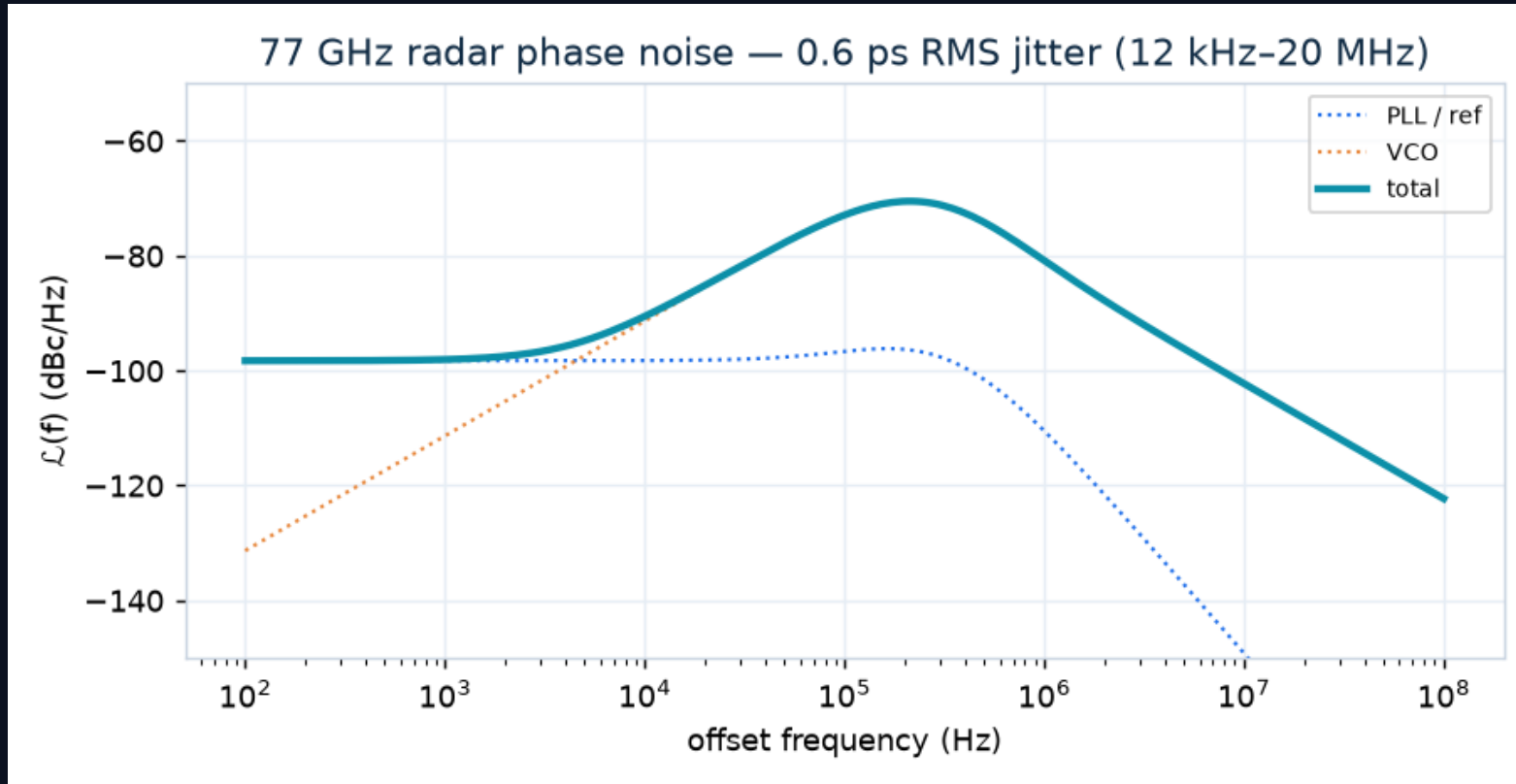


Full-stack + live

Control theory + RF/VLSI + ML + web, deployed at one URL on a safety-critical problem.

RESULTS

Phase noise & jitter at 77 GHz



77.0 GHz

radar RF

4.4 μ s

chirp settling

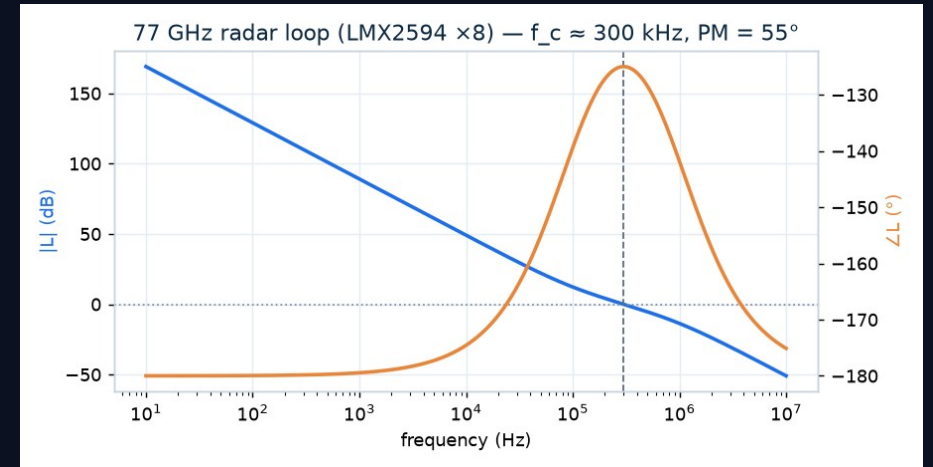
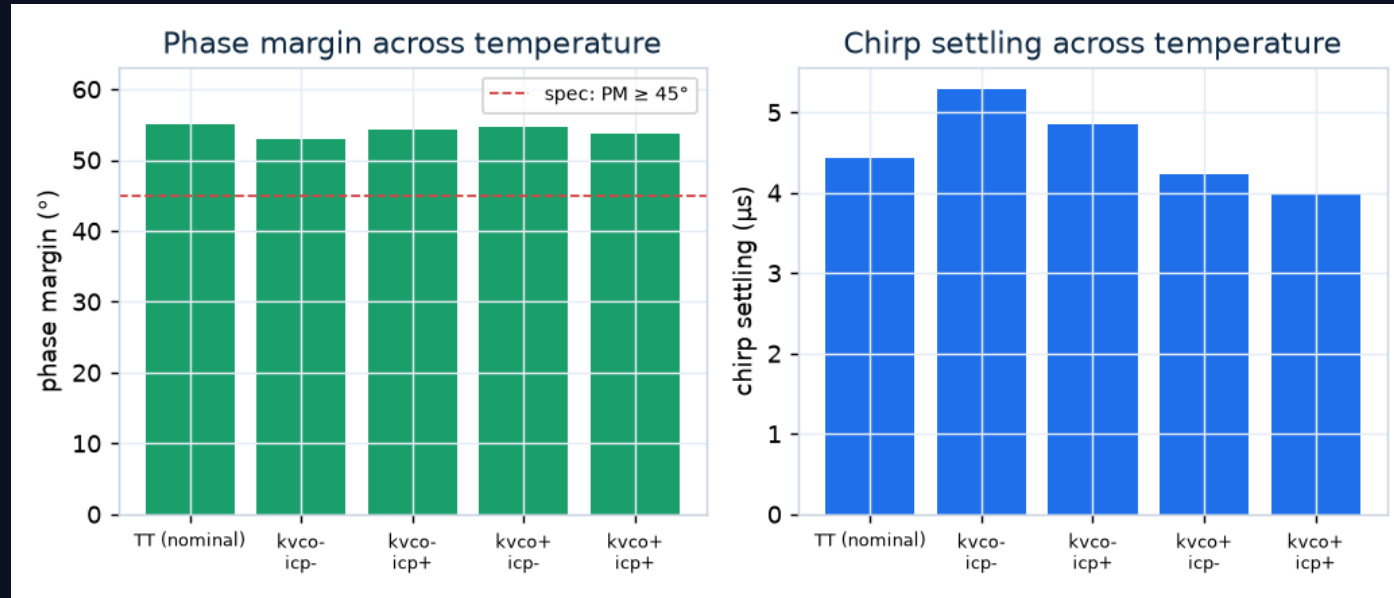
0.6 ps

RMS jitter

Real ADF/LMX-class 77 GHz radar profile · flat in-band, peaking bump at the loop bandwidth, VCO roll-off outside it.

RESULTS

Robust across the automotive temperature range



Phase margin stays above the 45° spec at every corner; the loop-filter zero gives the phase bump at crossover (right).

✓ Qualified across AEC-Q100 Grade 1 (–40 to +125 °C)

DEMO

The live web application

LockBench

DESIGNER

Design the **chip-synthesizer PLL** for an **AWW automotive radar** — and prove it holds chip settings, phase noise and jitter across the full automotive temperature range.

File in the table below (or load a preset), then press **Design & check temperature**. Finally, **Plot** and **Export** data as a CSV or PDF report.

1- Radar profile

reset

load a radar operating point

Design — the radar will be tested with a synthesizer loop operating point

CHIP SYNTHESIZER IC

LMX2594 — Texas Instruments

The test synthesizer chip is an LMX2594 PLL loop and its settings.

Mounted functional VCO synthesizer with integrated PLL

VCO 1: 0.000-10.000 MHz / VCO 2: 0.000-10.000 MHz / VCO 3: 0.000-10.000 MHz

2- Chip source — RF

VCO FREQ (Hz) MULTIPLIER (Hz)

9.425 8

about the synthesizer's operation, see the **chip-synthesizer** block in the **chip-synthesizer** block.

RF FREQ (Hz)

9.425

the synthesizer's output frequency, then the synthesizer must be used below 10.000 MHz.

3- Loop targets

LOOP BW (Hz) PHASE MARGIN (°)

300 55

about the loop's operation, see the **loop targets** block in the **loop targets** block.

CHARGE PUMP ICP (mA)

device default (0.25 mA)

Options — defaults to the chip's default current, higher up when the loop.

4- Temperature grade

AUTOMOTIVE QUALIFICATION (see PVT tab)

AEC-Q100 Grade 1 (-40-125 °C)

Options (default: AEC-Q100 Grade 1). The temperature range is the range of the device's operation, not the range of the chip's operation.

5- Pass/fail spec

The design must meet these at every temperature corner.

MIN PHASE MARGIN (°) MAX JITTER PERIOD (ns)

45 4

the design must meet these at every temperature corner. If it doesn't, the design is not valid.

Design & check temperature

Find robust design

ML recommend

Qualified across temperature

This loop filter holds every spec across AEC-Q100 Grade 1 (-40-125 °C).

RADAR RF

77.0 GHz

×8

CHIRP SETTling

4.43 μs

per chirp

PHASE NOISE

-81 dBc/Hz

@1 MHz, at RF

RMS JITTER

0.62 ps

12k-20M

Verdict + headline metric tiles

Designed loop filter

buildable R / C values · order 3

R

1.6 kΩ

C_MAIN

1.04 nF

C_SHUNT

115.25 pF

ZERO

94.59 kHz

3RD POLE

951.48 kHz

N (DIVIDER)

96

ICP

6.25 mA

KVCO

20 MHz/V

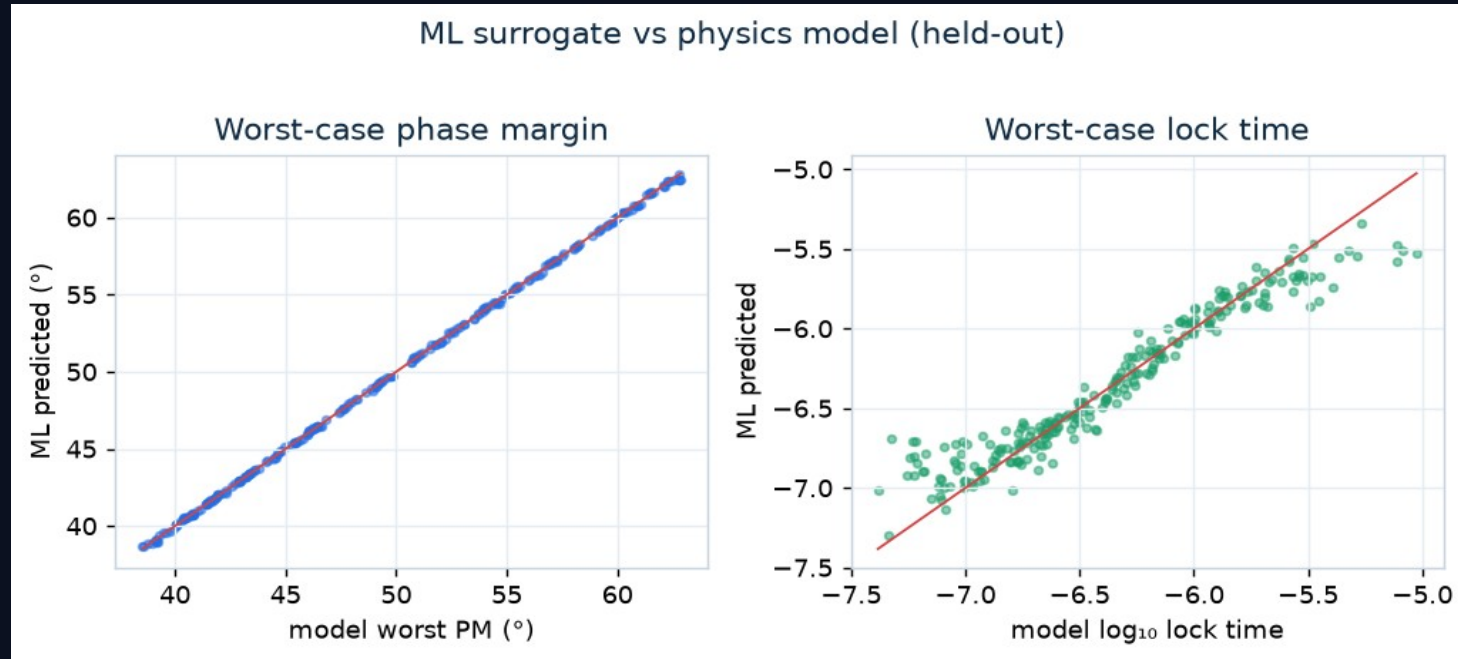
CHIRP CHAIN

×8 → 77.0 GHz

The full app

Buildable loop-filter output (R / C values)

Machine-learning inverse design



A random forest predicts a design's worst-case-across-PVT performance instantly.

So it scans hundreds of designs in milliseconds and recommends the best one from your spec.

It reproduces the physics model closely (left: phase margin; right: lock time) — and 'Apply' always re-runs the exact engine to verify.

How it is built — and its honest scope

Architecture

- React + Vite frontend (Plotly charts)
- FastAPI REST backend
- Python engine (NumPy, python-control)
- One Docker service on Render, one URL
- 62 tests, GitHub Actions CI

Scope & limitations

- Linear PLL model (valid for $BW \ll f_{PDF}$)
- Phase noise / jitter are first-order estimates
- Ideal $\times N$ multiplier to 77 GHz
- Temp grades $\rightarrow$ representative K_{vco}/I_{cp} drift
- ML ground truth is the model, not SPICE



Design → prove → deploy.

LockBench turns automotive-radar chirp-PLL loop-filter design into an instant, temperature-qualified, ML-accelerated workflow grounded in real device data.

References [1] F. M. Gardner, “Charge-Pump Phase-Lock Loops,” IEEE Trans. Communications, 1980. [2] “A Low-Phase-Noise 77-GHz FMCW Radar Transmitter with a 12.8-GHz PLL and ×6 Multiplier,” IEEE, 2016.